Anne Marie Jumper

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EDUCATION

Tufts University, Medford, MA — Graduating May 2027

B.S. Mechanical Engineering and Mathematics, GPA: 3.99, Dean's List (All Semesters)

- Awards and Honors:
  Tau Beta Pi Honor Society - 2025: Admitted due to scholastic record in top 1/8 of engineering class as a junior
  Howard Sample Prize in Physics - 2025: Awarded Tufts top student prize for performance in Physics I and II
- Relevant Coursework:
  Digital Control and Dynamic Systems, Thermal Fluid Systems I & II, Computational Fluid Dynamics, Electronics and Controls I & II, Nonlinear Dynamics and Chaos, AI in Mobile Robots, Senior Design, Introduction to Music Engineering, Instruments and Experiments, Mechanics I & II, Linear Algebra, Differential Equations, Partial Differential Equations, Multivariable Calculus
- Leadership & Involvement:
  President 2026, **Treasurer** 2024-2025. **Tufts American Society of Mechanical Engineers (ASME)**
  Teaching Assistant (TA) Mechanics II (Spring 2026) **and Thermal Fluid Systems II** (Fall 2026)
  Tufts Competitive Club Tennis Team. 2023 - Present. Advanced to Nationals in 2024/2025 season

WORK & RESEARCH EXPERIENCE

Research Fellow — Center for Engineering Education and Outreach, Tufts University — Summer 2026 - Present

- Develop and test experimental modal analysis features for ObieApp, in-house tool visualizing violin mode shapes and vibration data
- Perform FRF measurement and circle-fitting analysis on violin and aluminum reference plates, working to validate results against COMSOL and ANSYS FEA simulations and Instron material testing
- Investigate frequency effects of localized wood removed using magnet-based antinode detection, informing acoustic tuning techniques as part of ongoing senior thesis research
- Support integration of a UR robotic arm, guided by reinforcement learning, to automate impact testing for modal data collection

U.S. Department of Energy — Lawrence Berkeley National Laboratory (LBNL) — June - Aug 2025

Science Undergraduate Laboratory Intern (SULI*), Energy Storage & Distributed Resources Division*

- Contributed to an interdisciplinary team responsible for advancing solid-state battery (SSB) research and testing components through prototyping and computational modeling to improve reliability, lifetime, and EV integration.
- Modeled solid-state electrochemical cells in COMSOL Multiphysics to study current-density distribution in the electrodes, capturing insights to optimize SSB design, accelerate development and reduce risk of battery failure.
- Analyzed risk of dendrite growth at high current-density regions to identify optimal geometries for electrodes in SSBs.
- Fabricated graphite anodes, performing electrochemical cycling to evaluate stability and efficiency. Tested new procedures of coating spent graphite to develop affordable, easy to manufacture batteries using recycled elements.

ENGINEERING PROJECTS

Fluids Laboratory Projects - Computational Fluid Dynamics — Spring 2026

- Built a MacCormack-based finite-difference solver in **MATLAB** for the 1D unsteady compressible Navier-Stokes equations, modeling shock tube wave propagation and comparing 2nd- and 4th-order spatial schemes against the analytical solution
- Simulated 2D incompressible lid-driven cavity flow in **COMSOL Multiphysics** at low and high Reynolds numbers, analyzing vortex structure, boundary layers, and velocity and vorticity fields across viscous- and inertia-dominated regimes.

Multifunction Robot Design — Electronics and Controls I — Fall 2025

- Created a high-performing robot controlled by a microcontroller and **two Raspberry Pis** that became the benchmark for peer testing.
- Collaboratively designed and tested and independently built driving and payload transport components of the robot using **dual H-bridges** while measuring mass with **an amplifier-controlled load cell**.

Electric Ratchet Wrench Teardown — Engineering Design II — Fall 2025

- Computed speed, torque ratios, and tooth forces in stall conditions to observe device performance under worst-case loading and evaluate component design and materials. Synthesized results to inform product viability.
- Performed static stress simulation on the planetary gearbox using **SolidWorks** to identify product weaknesses

SKILLS

CAD and FEA: SolidWorks (modeling, drawing, FEA), COMSOL Multiphysics, ANSYS Fluent, Onshape, KiCAD
Software: MATLAB, LabVIEW, C++, Python, Java
Hardware & Fabrication: 3D printing, laser cutting, machining, electromechanical integration
Languages: English (Native), Spanish (Highly Proficient)